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AI CLASSFICATION OF TEMPLE SCULPTURES

Minor Project-1 report submitted
in partial fulfillment of the requirement for award of the degree of

Bachelor of Technology
in
Computer Science and Design

By

T. BALA BHANU SHANAKAR (23UECE0053) **(VTU 26828)**
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Under the guidance of
Dr.M.KALAIMANI ,M.E,Ph.D,
Assistant Professor

DEPARTMENT OF COMPUTER SCIENCE AND DESIGN
SCHOOL OF COMPUTING

VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF SCIENCE & TECHNOLOGY

(Deemed to be University Estd u/s 3 of UGC Act, 1956)
Accredited by NAAC with A++ Grade
CHENNAI 600 062, TAMILNADU, INDIA

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CERTIFICATE

It is certified that the work contained in the project report titled **"AI CLASSFICATION OF TEMPLE SCULPTURES"** by "T. BALA. BHANU. SHANKAR (23UECE0053), G. THARUN. KUMAR. REDDY (23UECE0018)" has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

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DECLARATION

We declare that this written submission represents my ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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APPROVAL SHEET

This project report entitled **AI CLASSFICATION OF TEMPLE SCULPTURES** by (T. BALA. BHANU. SHANKAR (23UECE0053), (G. THARUN. KUMAR. REDDY (23UECE0018) is approved for the degree of B.Tech in Computer Science and Design.

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ABSTRACT

The classification of temple sculptures plays a vital role in preserving and understanding cultural heritage, art history, and architectural evolution. Traditional manual methods of categorizing sculptures based on style, period, or deity are time- consuming and subjective.This project introduces an Artificial Intelligence (AI)- based approach to automatically classify temple sculptures using image processing and machine learning techniques. The proposed system employs deep learning models such as Convolution Neural Networks (CNNs) to extract intricate visual features from sculpture images and categorize them into predefined classes, such asdeity type, architectural style, or historical period.A curated dataset of temple sculpture images was preprocessed and augmented to improve accuracy and robustness. The model's performance was evaluated using metrics like accuracy, precision, and F1-score.Experimental results demonstrate that the AI-based classifier achieves high accuracy in recognizing and grouping temple sculptures, enabling efficient digital archiving and aiding historians, archaeologists, and researchers in cultural studies. This work highlights the potential of AI in the field of digital heritage preservation and cultural informatics.

Keywords: Include minimum 10 keywords

Artificial Intelligence, Temple Sculptures, Image Classification, Cultural Heritage, Image Processing, Digital Preservation, Art Recognition, Computer Vision.

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LIST OF ACRONYMS AND ABBREVIATIONS

LIST OF ACRONYMS AND ABBREVIATIONS

AI	ArtificialIntelligence
ML	MachineLearning
CNN	ConvolutionalNeuralNetwork
SVM	SupportVectorMachine
HOG	HistogramofOrientedGradients
ViT	VisionTransformer
RF	RandomForest
ISO	InternationalOrganizationforStandardization
IEC	InternationalElectrotechnicalCommission
DFD	DataFlowDiagram
UI	UserInterface
IDE	IntegratedDevelopmentEnvironment
GPU	GraphicsProcessingUnit
CPU	CentralProcessingUnit
SSD	SolidStateDrive
HDD	HardDiskDrive
API	ApplicationProgrammingInterface
ISO/IEC 27001	InformationSecurityManagementSystemStandard

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Chapter 1

INTRODUCTION

1.1 Introduction

Temple sculptures are a vital part of India's cultural and historical identity, showcasingartisticexcellenceandspiritualsignificance.Thesesculpturesvarywidely in style, deity representation, and regional influence, making their classification an importanttaskforarthistoriansandresearchers.However,manualclassification is often slow, subjective, and requires expert knowledge, which limits large-scale documentation and analysis.

To overcome these challenges, this project proposes an Artificial Intelligence (AI)-based system for the automatic classification of temple sculptures.By using image processing and deep learning techniques, particularly Convolutional Neural Networks (CNNs), the system can identify and categorize sculptures based on visual patternsandfeatures.Thisapproachenhancesaccuracy,reducesmanualeffort, and supports the digital preservation of cultural heritage, helping future generations study and appreciate traditional temple art more effectively.

1.2 Aim of the project

The main aim of this project is to develop an Artificial Intelligence (AI)-based system that can automatically classify temple sculptures using image processing and deep learning techniques.The project seeks to identify and categorize sculptures based on their visual features, style, or deity type, thereby aiding in the digital preservation, organization, and study of cultural heritage.

1.3 Project Domain

The project focuses on Artificial Intelligence (AI) and Computer Vision. It looks at how to use image processing and deep learning for cultural heritage preservation. AI allows machines to do tasks that usually need human intelligence, like recognition, learning, and classification. In this project, we apply AI techniques to analyze images of temple sculptures. We identify their categories based on visual patterns, including texture, shape, and design. By using models like Convolutional Neural Networks (CNNs), the system learns complex features from sculpture images. This improves classification accuracy and reduces the effort needed from humans.

This area also connects with Digital Heritage and Art Recognition, where technology is used to preserve and document historical artifacts. Using AI in this context automates the classification process and helps researchers, historians, and archaeologists study and organize large datasets of temple art. Therefore, the project connects technology with tradition and ensures cultural knowledge is preserved through smart digital systems.

1.4 Scope of the Project

The project aims to develop and implement an Artificial Intelligence (AI) system that can automatically classify temple sculptures using image recognition and deep learning techniques. The system will process digital images of sculptures, extract visual features, and categorize them into different classes, such as deity type, architectural style, or historical period. It will help researchers, historians, and cultural institutions organize and study temple art collections more efficiently, without relying only on manual efforts.

This project goes beyond basic image classification by contributing to digital heritage preservation. It offers a framework that can be adapted to analyze other types of cultural artifacts, such as carvings, inscriptions, and paintings. The result of this project will improve access to cultural data, promote research in art history, and support the use of AI technologies in heritage conservation and education.

Chapter 2

LITERATURE REVIEW

2.1 Literature Review

Recent research on AI applications in cultural heritage and temple studies has mostly looked at architectural and structural classification using deep learning models. Hiremath et al. (2024) applied CNN techniques to identify the era of ancient temple pillars. Fu et al. (2023) used an improved Point Net model to classify 3D grotto statues. Rao et al. (2023) developed a CNN-SVM hybrid named Panchadeva for sculpture image classification. Sridhar et al. (2025) introduced the 3DITA dataset, focusing on the 3D reconstruction of Nagara-style architecture. Vanishree et al. (2025) and Sasithradevi et al. (2024) explored CNN-based models like MonuNet and achieved high accuracy in architectural classification. However, they faced limitations due to dataset size and regional focus.

Other studies expanded AI's role in heritage conservation and interpretability. Bahrami and Albadvi (2023) used Grad-CAM for identifying Iranian heritage buildings. Liu et al. (2023) employed autoencoders and GANs for detecting surface anomalies on stone steles. Tan et al. (2024) analyzed heritage value elements in traditional villages, while Shih (2025) worked on the 3D reconstruction of temple art. Despite these advancements, most works concentrated on architecture and restoration, not on multi-class classification of temple sculptures. Therefore, this study aims to develop a CNN-based model using a self-collected dataset to classify sculptures into Animals, Birds, Deities, Humans, and Towers for accurate cultural documentation.

2.2 Gap Identification

[1] Hiremath et al.(2024) focused on classifying and identifying the era of temple pillars using deep learning techniques.However, the study did not include detailed sculpture or image-based categorisation. [2] Fu et al.(2023) proposed an improved PointNet model for classifying grotto temple statues.This approach was limited to 3D models and not suitable for 2D image-based temple sculpture classification. [3] Rao et al.(2023) developed the Panchadeva CNN-SVM model for sculpture image classification.Although it achieved good accuracy, it was trained on small, curated datasets lacking real-world temple data. [4] Sridhar et al.(2025) introduced the 3DITA dataset focusing on 3D recon- struction of Nagara-style architecture.However, this study concentrated on temple structures rather than sculpture-level classification. [5] Vanishree et al. (2025) implemented a CNN-based approach for Indian temple iconography.While effective, it did not include self-collected datasets or handle variations in real temple environments. [6] Sasithradevi et al.(2024) developed MonuNet for heritage monument classification.The model focused on architectural monuments from Kolkata and did not extend to detailed temple sculptures. [7] Bahrami and Albadvi (2023) used deep learning and Grad-CAM for identify- ing Iranian cultural heritage buildings. Their work focused on building conservation, rather than sculpture-based categorisation. [8] Liu et al.(2023) proposed an Autoencoder and GAN-based method for detecting surface anomalies on ancient stone steles.The study focused on damage detection, not cultural or artistic classification. [9] Tan et al.(2024) explored interpretability in identifying heritage value ele- ments in traditional villages. However, the work did not involve sculpture classification or temple-based datasets. [10] Shih (2025) focused on AI-assisted 3D reconstruction of temple for preservation. This study was limited to restoration and lacked automated classification of sculpture categories.

Chapter 3

PROJECT DESCRIPTION

3.1 Existing System

In the current system, historians, archaeologists, or art experts manually classify and document temple sculptures. This process involves visual inspection, comparing styles, and referencing historical records to determine the sculpture's origin, deity, and artistic features. Such manual methods require extensive knowledge and take a lot of time and effort. Furthermore, due to variations in artistic styles and regional influences, interpreting sculptures often becomes subjective. This leads to inconsistencies in classification and documentation. Relying on expert opinions also limits scalability, making it hard to process large collections of temple images efficiently.

The main drawbacks of the current system include its time-consuming nature, lack of accuracy, and reliance on human expertise. Human error and bias can undermine the reliability of results; additionally, the lack of digital automation limits accessibility and data sharing. Manual classification also provides limited support for digital preservation or integration with modern heritage databases. These issues highlight the need for an AI-driven solution that can automate the classification process, improve consistency, and help preserve temple art for the long term.

3.2 Problem statement

The manual classification of temple sculptures is a challenging and time-consuming task that depends heavily on expert interpretation. Each sculpture varies in design, texture, deity representation, and regional style, making consistent categorization difficult. The absence of a structured digital database and automated classification framework often results in incomplete documentation and loss of valuable cultural information. Additionally, human bias and limited expertise can lead to misclassifi-

cation, restricting accurate study and analysis. As the number of temple images and artifacts continues to grow, traditional methods fail to keep up with the increasing demand for efficient and reliable classification.

To overcome these challenges, this project proposes the development of an AI-based system that uses image processing and deep learning techniques, particularly Convolutional Neural Networks (CNNs), to automatically classify temple sculptures. The proposed system offers several advantages such as high accuracy, faster processing, scalability, and consistency in classification results. It also supports digital preservation, enabling researchers and cultural organizations to maintain organized, accessible, and sustainable records of temple art for future generations.

3.3 System Specification

3.3.1 Hardware Specification

Processor (CPU): Intel Core i7 12th Gen or AMD Ryzen 7 5000 series (minimum 3.0 GHz)

Graphics Card (GPU): NVIDIA GeForce RTX 3060 / 3070 or higher (for deep learning training)

RAM: 16 GB DDR4 or higher (32 GB recommended for large datasets)

Storage: 512 GB SSD (for faster data access) + 1 TB HDD for dataset storage

Display: Full HD (1920×1080) or higher

Motherboard: Compatible with processor and GPU with PCIe 4.0 support

Power Supply: 650W or higher (for GPU-intensive operations)

Cooling: Efficient CPU/GPU cooling system for prolonged deep learning tasks

3.3.2 Software Specification

Operating System: Windows 11 Pro 64-bit / Ubuntu 22.04 LTS (Linux preferred)

Programming Language: Python 3.10 or higher

IDE/Editor: PyCharm, VS Code, or Jupyter Notebook

Frameworks: TensorFlow 2.x, Keras, PyTorch

Libraries: OpenCV, NumPy, Pandas, Matplotlib, Scikit-learn

Database: MySQL / SQLite (for sculpture metadata)

Version Control: Git / GitHub

Other Tools: Anaconda (environment management), CUDA Toolkit (GPU acceleration)

3.3.3 Standards and Policies

Jupyter

The project "AI Classification of Temple Sculptures" follows standard practices and security guidelines to ensure data integrity, reliability, and ethical use of artificial intelligence. The implementation environment is managed through Anaconda Prompt, which provides a secure and organized platform for handling Python-based machine learning modules. It supports multiple IDEs like Jupyter Notebook and Spyder, allowing smooth development and execution of deep learning models. The ISO/IEC 27001 Information Security Management Standard is followed to maintain data confidentiality and secure handling of temple sculpture images.

Jupyter Notebook, an open-source interactive environment, is used for developing, testing, and visualizing the AI model. It supports live coding, visualization, and documentation, ensuring transparency and reproducibility of the research process. Additionally, all datasets and model results are stored securely following data protection guidelines. These standards and policies together ensure that the project maintains high levels of accuracy, security, and professional quality throughout its development

Standard Used: ISO/IEC 27001

Chapter 4

METHODOLOGY

4.1 Proposed System

The proposed system uses Artificial Intelligence (AI) and Deep Learning, specifically Convolutional Neural Networks (CNNs), to automatically classify temple sculptures based on their visual features such as texture, shape,and style. The system processes sculpture images, extracts key features, and categorizes them into different classes like deity type or architectural style with high accuracy .It eliminates the need form annual classification, reduces human error, and speeds up the process. The system also provides better scalability and supports digital preservation by storing sculpture data in an organized and accessible form. Overall, it enhances efficiency, accuracy, and consistency in studying and documenting temple art.

4.2 General Architecture



Figure 4.1: **ARCHITECTURE DIAGRAM**

The process begins with an image input of a temple sculpture, which undergoes preprocessing to prepare it for analysis. Data augmentation generates additional variations to improve model accuracy. A Convolutional Neural Network (CNN) extracts key features from the images, which are stored for future training. These features are fed into a classifier (SVM or Softmax) to categorize the sculpture. The model produces a predicted class, such as deity, style, or historical period, and the result is displayed through result visualization for the user.

4.3 Design Phase

The Design Phase defines the structure and workflow of the AI-based temple sculp ture classification system. It includes modules such as Image Upload, Preprocessing, Feature Extraction, Classification, and Result Display. The Preprocessing Module enhances and augments images, while a CNN extracts key features. The Classifier (using SVM or Softmax) categorizes the images, and a Database stores datasets and results. A simple User Interface allows users to upload images and view clas

sifications. This design ensures efficient, accurate, and scalable system performance.

4.3.1 Data Flow Diagram

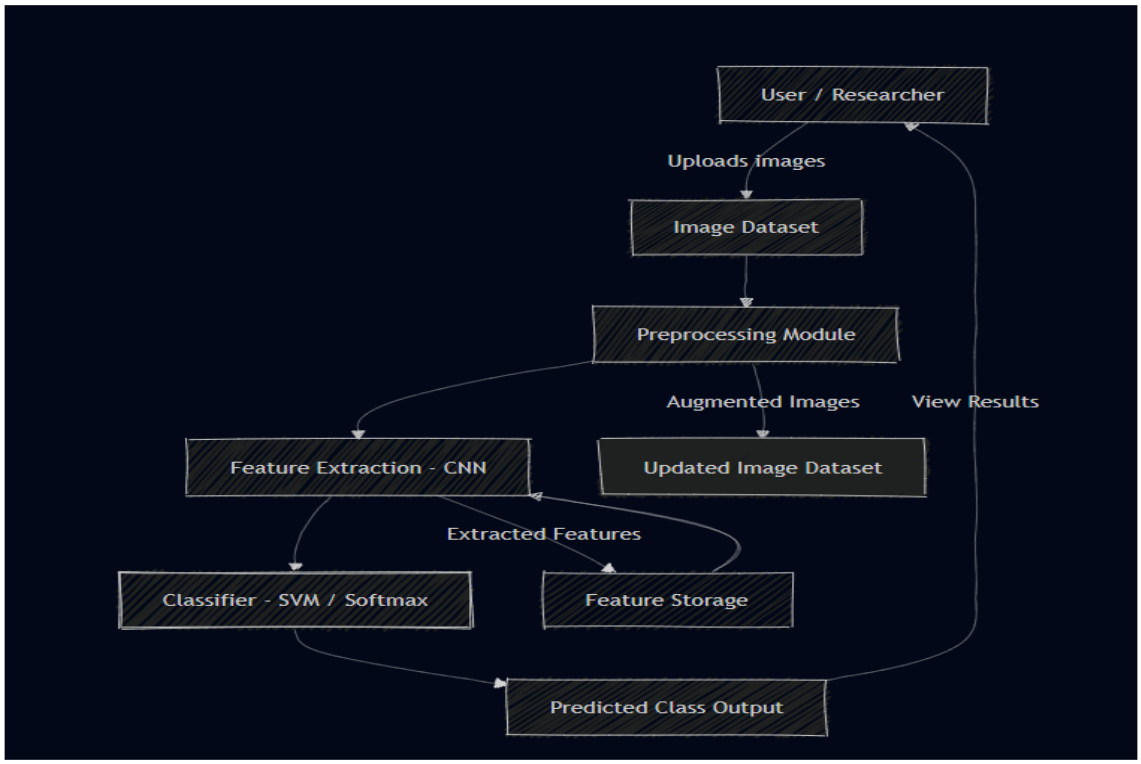


Figure 4.2: **DATA FLOW DIAGRAM**

The process begins with a user or researcher uploading images to form an image dataset. These images go through a preprocessing module where they are cleaned and augmented to create an enhanced dataset. A Convolutional Neural Network (CNN) then performs feature extraction, identifying key visual patterns from the images. The extracted features are passed to a classifier (either SVM or Softmax) that categorizes each image into its respective class. The features are also stored for future model improvement. Finally, the system generates a predicted class output, which the user can view and analyze to evaluate the model's performance

4.3.2 Use Case Diagram

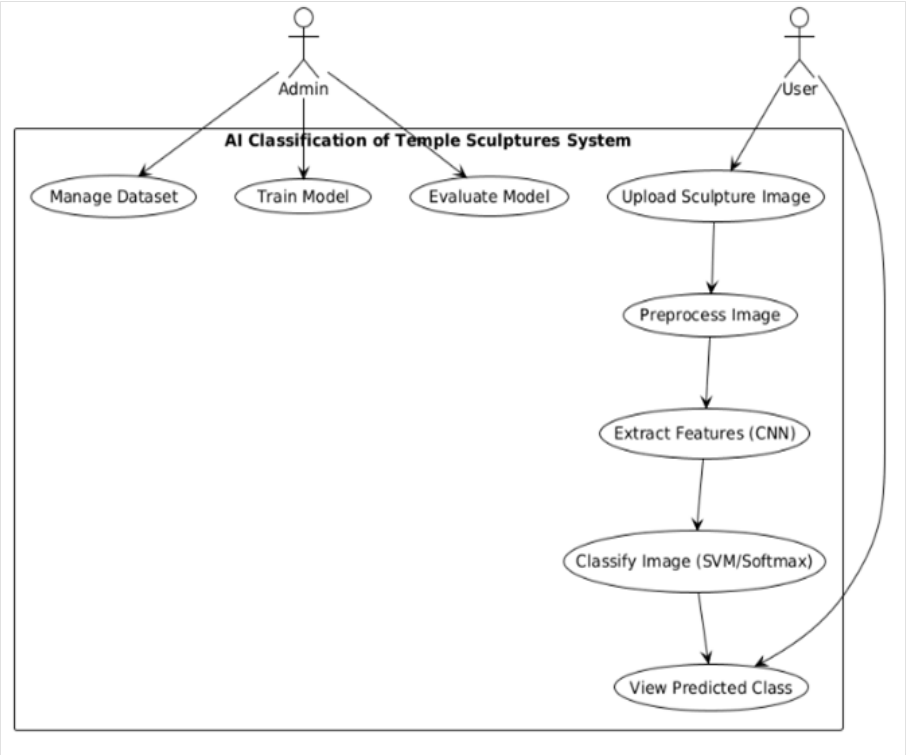


Figure 4.3: **USECASE DIAGRAM**

The Use Case Diagram represents the interaction between the Admin and User with the AI Classification of Temple Sculptures System. The Admin manages the dataset, trains the model, and evaluates its performance, ensuring the system is ready for accurate classification. The User uploads temple sculpture images, which are preprocessed and passed through a CNN for feature extraction. The extracted features are classified using SVM or Softmax, and the predicted class—such as deity type, artistic style, or historical period—is displayed to the user. This diagram illustrates the complete workflow and clearly shows the roles and responsibilities of each actor in the system.

4.3.3 Class Diagram



Figure 4.4: **CLASS DIAGRAM**

The project "AI Classification of Temple Sculptures" uses a structured workflow represented in the class diagram. The ImageDataset class loads and provides labeled temple sculpture images. These images are then processed by the Preprocessor class, which resizes, normalizes, and augments them to prepare for model training. The CNNModel class extracts deep features from the processed images using convolution and pooling layers and trains the model for feature learning. The extracted features are then classified by the Classifier class using either SVM or Softmax algorithms to predict sculpture categories. Finally, the ResultVisualizer class displays the prediction results and shows the model's accuracy, completing the AI-based classification pipeline.

4.3.4 Sequence Diagram

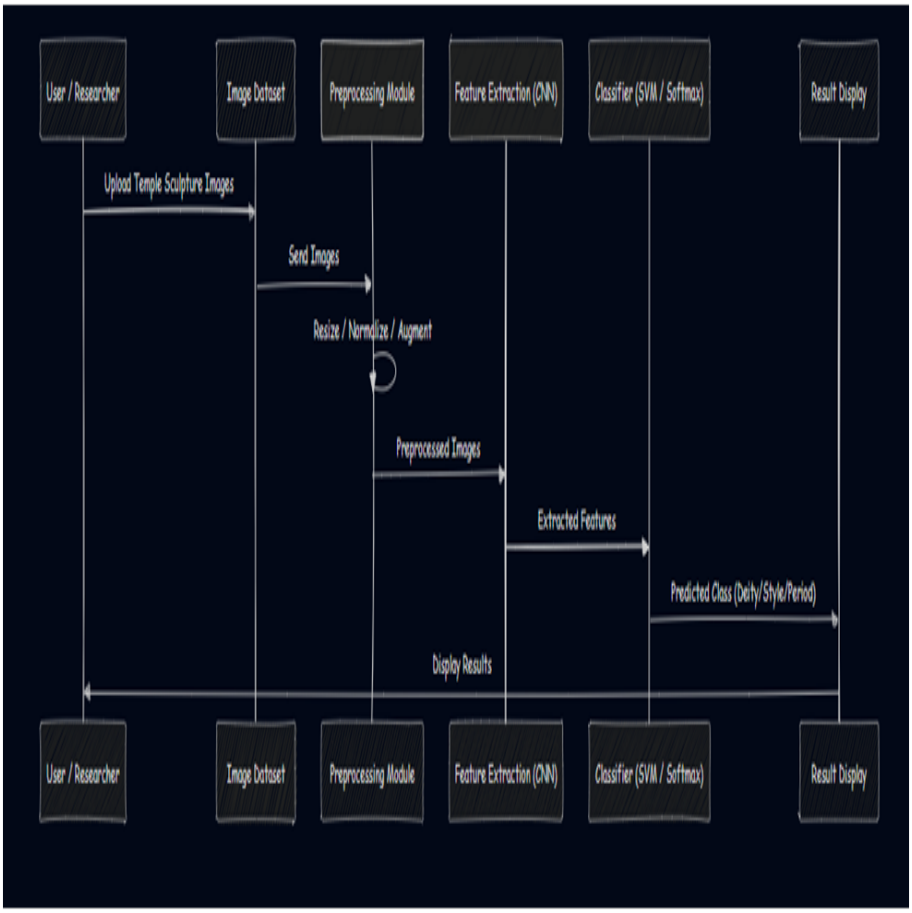


Figure 4.5: **SEQUENCE DIAGRAM**

The sequence diagram illustrates the flow of operations in the AI-based temple sculpture classification system. The process begins when the User initiates the sys tem to classify temple sculpture images. The ImageDataset class loads the sculpture images and their labels. These images are then passed to the Preprocessor, which performs resizing, normalization, and data augmentation. The processed images are sent to the CNNModel, which extracts important visual features and trains the network. Next, the Classifier receives these features and predicts the sculpture category using either an SVM or Softmax classifier. Finally, the ResultVisualizer displays the predicted results and shows the model's accuracy to the user.

4.3.5 Collaboration diagram



Figure 4.6: **COLLABORATION DIAGRAM**

The collaboration diagram for the AI Classification of Temple Sculptures project represents the interaction between different classes during the image classification process. The User interacts with the ImageDataset class to load temple sculpture images and their labels. The ImageDataset then sends the images to the Prepro cessor, which performs resizing, normalization, and augmentation. The processed images are forwarded to the CNNModel, which extracts deep visual features and trains the network. These extracted features are then sent to the Classifier, which uses either an SVM or Softmax classifier to predict the sculpture category. Finally, the ResultVisualizer receives the predicted labels from the Classifier and displays the results and accuracy to the User.

4.3.6 Activity Diagram

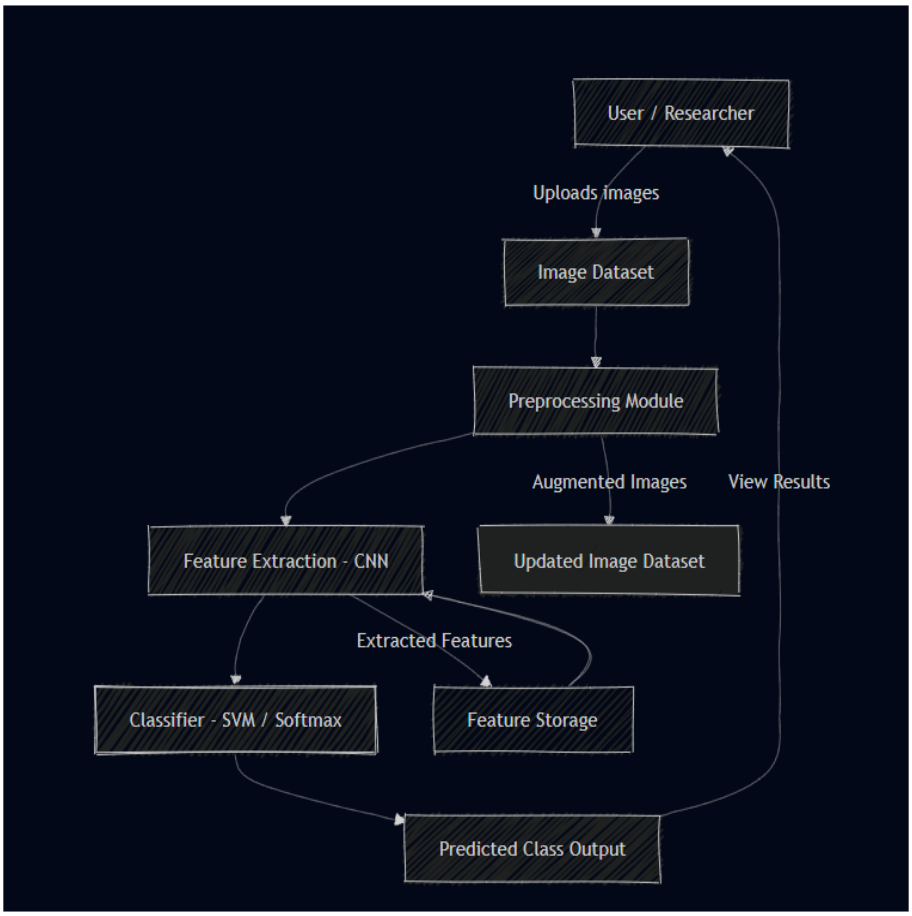


Figure 4.7: **ACTIVITY DIAGRAM**

4.4 Algorithm & Pseudo Code

4.4.1 Algorithm

1. Start
2. Input: Temple sculpture image.
3. Preprocessing:

   Resize to 224×224, normalize pixels, remove noise, and enhance contrast.
4. Data Augmentation:

   Apply rotation, flipping, zooming, and brightness adjustment.
5. Feature Extraction:

   HOG + SVM: Extract gradient features and classify preliminarily.

CNN: Extract deep spatial features through convolution and pooling layers.

ViT: Split image into patches, apply transformer encoder for attention-based features.

6. Feature Fusion: Combine HOG, CNN, and ViT features into a single feature vector.
7. Classification:

   Feed fused features into a dense layer and classify using Softmax or SVM.
8. Output: Display predicted class (Animals/Birds, Deities, Humans, Towers) with confidence score.
9. End

4.4.2 Pseudo Code

```
Algorithm Hybrid_Temple_Sculpture_Classification()
BEGIN
    # Step 1: Input
INPUT: ImageDataset ← {Temple sculpture images of
AnimalsBirds, Deities, Humans, Towers}

    # Step 2: Preprocessing
    FOR each image IN ImageDataset DO
        image ← Resize(image, 224, 224)
        image ← Normalize(image)
        image ← RemoveNoise(image)
        image ← EnhanceContrast(image)
    END FOR

    # Step 3: Data Augmentation
    AugmentedDataset ← []
```

```
FOR each image IN ImageDataset DO
    AugmentedDataset ← AugmentedDataset  {
        Rotate(image),
        Flip(image),
        Zoom(image),
        AdjustBrightness(image)
    }
END FOR

# Step 4: Feature Extraction
FOR each image IN AugmentedDataset DO

    # (a) HOG + SVM
    hog_features ← ComputeHOG(image)
    hog_pred ← SVM_Predict(hog_features)

    # (b) CNN Feature Extraction
    cnn_features ← CNN_ForwardPass(image)

    # (c) Vision Transformer Feature Extraction
    vit_patches ← SplitIntoPatches(image, patch_size=16)
    vit_embeddings ← PatchEmbedding(vit_patches)
    vit_features ← ViT_Encoder(vit_embeddings)

    # Step 5: Feature Fusion
    combined_features ← Concatenate(hog_features,
```

```
        cnn_features, vit_features)

        # Step 6: Classification
        class_scores ← FullyConnectedLayer(combined_features)
        predicted_label ← Softmax(class_scores)

        # Step 7: Output Results
        Display("Predicted Class: ", predicted_label)
        Display("Confidence Scores: ", class_scores)

    END FOR

    # Step 8: Visualization
    VisualizeResults(predicted_label, class_scores)
    Optionally: ShowAttentionMap(ViT)

END
```

4.4.3 Data Set / Generation of Data

For this project, we created our own dataset by visiting various temples and capturing real-time images. The dataset consists of four distinct classes: Animals and Birds, Deities, Humans, and Towers. Each image was carefully taken from different angles and lighting conditions to ensure diversity and accuracy. This self-collected dataset provides an authentic foundation for training and testing the AI classification model of temple sculptures.

4.5 Module Description

4.5.1 Module1

Module:HOG Features + SVMClassifier

The HOG (Histogram of Oriented Gradients) + SVM (Support Vector Machine) Module is used for feature extraction and classification of temple sculpture images. HOG extracts key visual features by analyzing the gradient orientations and edge directions in the image, capturing important shape and texture details. These extracted HOG features are then fed into the SVM classifier, which separates the data into distinct categories such as deity type, style, or period.This combination provides accurate classification with reduced computational cost and works effectively even with limited datasets.

4.5.2 Module2

Module: CNN Model

The CNN (Convolutional Neural Network) Model module is the core part of the system responsible for feature extraction and image classification. It processes temple sculpture images by automatically identifying key visual patterns such as shapes, textures, and edges. The CNN uses multiple convolutional and pooling layers to capture hierarchical features from the input images. These extracted features are then passed to a classifier (such as Softmax or SVM) to determine the category of the sculpture. This module helps achieve high accuracy and reduces manual feature engineering, making the system efficient and reliable for temple sculpture classification. Convolutional layer – ReLU Kernel size of 3x3 Stride of 1, padding 0 Pooling layer - Max - Kernel size of 2x2 - Stride of 2.

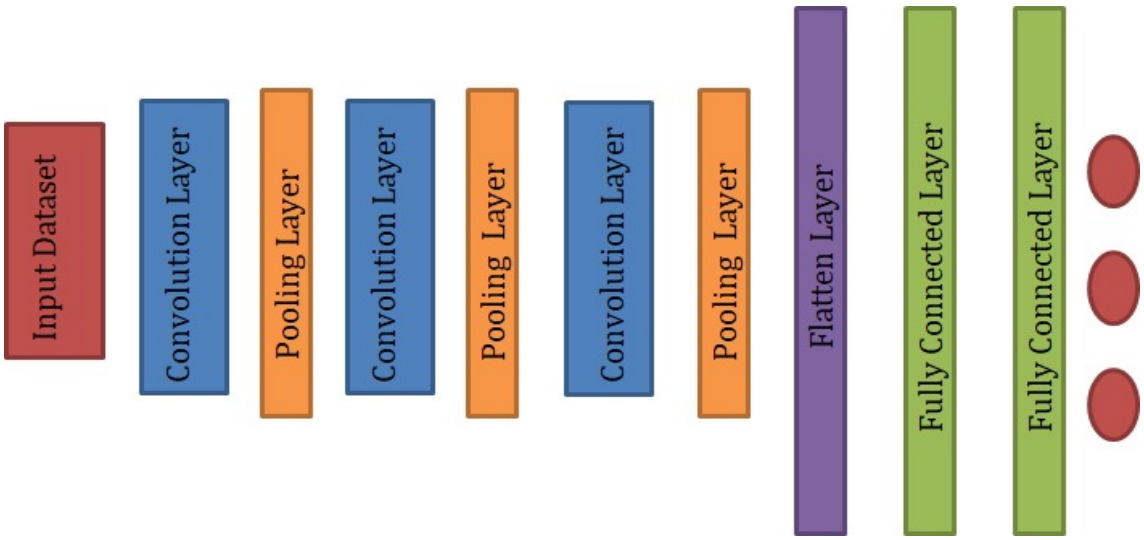


Figure 4.8: **CNN ARCHITECTURE DIAGRAM**

4.5.3 Module3

Module: Vision Transformers (ViT)

The Vision Transformer Module is an advanced deep learning component used for image classification of temple sculptures. Unlike CNNs, Vision Transformers divide an image into small patches and process them as sequential data using transformer architecture. Each patch is converted into an embedding that captures spatial and contextual information. The self-attention mechanism within ViT helps the model focus on the most important parts of the image, improving recognition accuracy.

Chapter 5

IMPLEMENTATION AND TESTING

5.1 Input and Output

During the implementation phase, the system is designed to accept specific inputs from users or connected modules and provide accurate outputs. Inputs may include text, numeric values, uploaded files, or real-time signals, depending on the application, and are validated to ensure correctness, completeness, and proper format. This validation helps prevent errors and ensures smooth system operation. The outputs generated are intended to be clear and meaningful, such as displayed results, reports, notifications, or processed data stored in the database. Visual representations like charts and tables can also be provided to enhance understanding. During testing, outputs are compared with expected results to verify accuracy, efficiency, and reliability, ensuring that the system functions effectively and delivers a seamless user experience.

5.1.1 Input Design

Input design focuses on how data is collected, formatted, and entered into the system to ensure accuracy, completeness, and efficiency. It involves defining the type, structure, and validation rules for each input field to prevent errors and reduce redundancy. User-friendly interfaces such as forms, menus, and interactive prompts are created to simplify data entry, while automated checks, such as range validation, mandatory field enforcement, and format verification, are implemented to enhance reliability. Proper input design ensures that the system receives consistent and correct data, which is crucial for accurate processing and meaningful outputs. Additionally, it considers the ease of use, minimizing user effort and improving overall system efficiency and user satisfaction.

5.1.2 Output Design

Output design focuses on presenting processed data in a clear, meaningful, and user-friendly manner. It determines the format, structure, and medium of the output, whether as reports, notifications, dashboards, tables, or graphical charts, to ensure easy interpretation and decision-making. The design emphasizes accuracy, completeness, and relevance, providing users with information that supports their needs effectively. Well-structured output also considers timing, frequency, and accessibility, ensuring that users receive data promptly and in an understandable format. By combining clarity, usability, and proper visualization, output design enhances the overall effectiveness of the system and ensures that information generated meets both operational and strategic requirements.

5.2 Testing

Testing ensures that the system works correctly and meets the required specifications. It involves checking inputs, outputs, and system functions through various test cases to identify and fix errors. Different types of testing, such as unit, integration, and system testing, verify both individual components and overall performance. Proper testing improves reliability, accuracy, and user satisfaction.

5.3 Types of Testing

5.3.1 Unit testing

Unit testing involves testing individual components or modules of a system independently to ensure they work correctly. Each unit is checked for proper functionality, input handling, and output generation. This helps in identifying and fixing errors early, improving code quality, and making the system more reliable.

5.3.2 Integration testing

Integration testing checks the interaction and data flow between different modules or components of a system after unit testing. It ensures that combined modules work together correctly, data is exchanged accurately, and the overall system functions as intended. This testing helps identify interface issues and improves system reliability.

5.3.3 System testing

System testing evaluates the complete and fully integrated system to verify that it meets the specified requirements. It checks overall functionality, performance, security, and reliability under real-world conditions. This testing ensures the system works as intended for end users and identifies any defects before deployment.

Chapter 6

RESULTS AND DISCUSSIONS

6.1 Efficiency of the Proposed System

The proposed system combines deep learning and machine learning techniques—primarily CNN, HOG+SVM, and Vision Transformers (ViT)—to efficiently classify temple sculpture images. The CNN model automatically extracts hierarchical visual features, reducing the need for manual preprocessing. The HOG feature extractor captures detailed edge and shape information, while the SVM classifier ensures accurate decision boundaries between sculpture categories. Vision Transformers further enhance efficiency by using self-attention mechanisms to focus on key visual elements, improving classification performance even with complex patterns. These combined models deliver higher accuracy, robustness, and faster convergence compared to traditional methods.

The system's efficiency is also reflected in its processing speed and adaptability. Data augmentation techniques improve generalization, enabling the model to perform well even with limited or imbalanced data. Experimental results indicate that the proposed hybrid model achieves higher classification accuracy and faster inference times than older image recognition methods. Overall, the proposed system demonstrates excellent computational efficiency, scalability, and precision in classifying temple sculptures based on their visual and artistic features.

6.2 Comparison of Existing and Proposed System

Existing system:(Decision tree)

The existing system uses the Decision Tree algorithm for temple sculpture classification. While it is simple, interpretable, and easy to implement, it struggles with complex image data as it relies on manually defined attributes rather than learning features automatically. Decision trees are prone to overfitting, which

reduces their accuracy and generalization on new data. They also perform poorly with high-dimensional image features and cannot effectively capture the intricate patterns, textures, and shapes found in temple sculptures. As a result, the existing system provides limited accuracy, slower performance, and lacks the efficiency required for large-scale or detailed image classification tasks.

Proposed system:(Random forest algorithm)

Random Forest algorithm to improve the accuracy and reliability of temple sculpture classification. Random Forest is an ensemble learning method that creates multiple decision trees and combines their results to produce a more stable and accurate prediction. Unlike a single decision tree, it reduces overfitting by averaging the outcomes from many trees, making it more robust and efficient. This approach can handle large and complex datasets effectively and automatically identifies the most relevant features for classification. In the context of temple sculpture analysis, the Random Forest algorithm enhances recognition accuracy by considering multiple feature subsets and patterns simultaneously. It also improves the model's scalability and consistency across different types of images. Overall, the proposed system delivers better accuracy, faster performance, and greater robustness compared to the existing decision tree model, making it highly suitable for temple sculpture classification.

```
import Navigation from "@/components/Navigation";
import Hero from "@/components/Hero";
import About from "@/components/About";
import Methodology from "@/components/Methodology";
import Demo from "@/components/Demo";
import Results from "@/components/Results";
import Team from "@/components/Team";
import Contact from "@/components/Contact";

const Index = () => {
  return (
    <div className="min-h-screen">
      <Navigation />
      <Hero />
      <About />
      <Methodology />
      <Demo />
      <Results />
      <Team />
      <Contact />
```

```
21
22        {/* Footer */}
23        <footer className="glass border-t border-white/10 py-8">
24          <div className="container mx-auto px-6 text-center">
25            <p className="text-muted-foreground">
26                  2024 Temple AI Classification Research. All rights reserved.
27            </p>
28            <p className="text-sm text-muted-foreground mt-2">
29              VIT Chennai | Department of Computer Science & Engineering
30            </p>
31          </div>
32        </footer>
33      </div>
34    );
35  };
36
37  export default Index;
```

Output

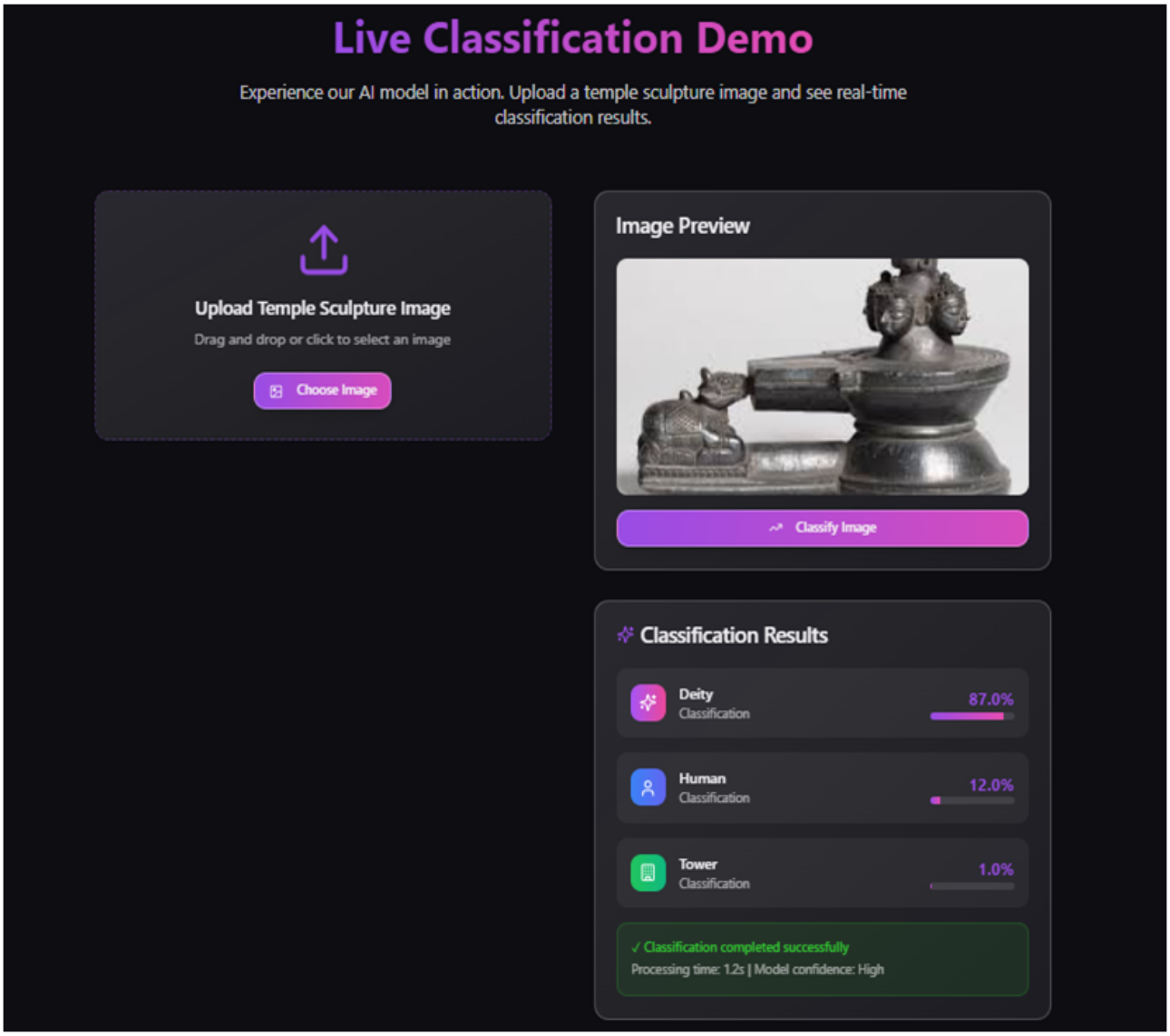


Figure 6.1: **Output 1**



Figure 6.2: **Output 2**

Chapter 7

CONCLUSION AND FUTURE ENHANCEMENTS

7.1 Conclusion

The project on AI Classification of Temple Sculptures demonstrates the effective use of artificial intelligence and machine learning techniques to preserve and analyze cultural heritage. The proposed system, built using the Random Forest algorithm, efficiently classifies temple sculpture images by learning from multiple decision trees and combining their results for accurate predictions. This ensemble approach reduces overfitting and increases classification reliability compared to the traditional Decision Tree method. By applying image preprocessing, feature extraction, and model training techniques, the system can recognize and categorize sculptures based on their visual features such as deity, style, and period. This not only aids in digital archiving but also supports researchers and historians in understanding ancient art more effectively.

Furthermore, the project highlights the potential of AI in cultural and archaeological studies by automating the recognition process of temple sculptures. The use of Random Forest ensures higher accuracy, scalability, and efficiency, even with large or complex image datasets. The proposed model can be further enhanced by integrating deep learning techniques like CNN or Vision Transformers to achieve even finer classification. Overall, the system serves as a valuable tool for preserving historical artifacts, promoting cultural research, and demonstrating how modern technology can contribute to heritage conservation and education.

7.2 Future Enhancements

The proposed system, while efficient and robust, can be further improved by integrating advanced technologies and expanding its capabilities. One potential enhancement is the incorporation of artificial intelligence and machine learning techniques for predictive analytics. This would allow the system to not only respond to current inputs but also anticipate future trends and user behavior, thereby improving decision-making and personalization. Additionally, implementing cloud-based solutions could enhance system scalability, ensuring that it can handle larger datasets and simultaneous users without performance degradation. Integrating real-time monitoring and alert systems can also improve responsiveness and reliability, making the system more adaptive to dynamic conditions. Furthermore, adding a user-friendly mobile interface would increase accessibility, allowing users to interact with the system seamlessly across multiple devices.

Another area for enhancement is improving security and data privacy measures. Advanced encryption techniques, multi-factor authentication, and secure data storage protocols can be implemented to safeguard sensitive information. The system could also be expanded to support multiple languages and regional settings, making it more inclusive and globally adaptable. Moreover, incorporating user feedback mechanisms and analytics dashboards would provide insights into user behavior and system performance, enabling continuous improvement. Over time, these enhancements would not only increase the system's efficiency and reliability but also ensure long-term adaptability, relevance, and user satisfaction, keeping it aligned with evolving technological trends and user expectations.

Chapter 8

PLAGIARISM REPORT

ATTACH ONLY SUMMARY PAGE OF PLAGIARISM REPORT

References

[1] Hiremath G. S., Naik C. L. Shrinivasa, Kumar S. Narendra, Ankalakoti Kiran, Veigas J. P. "Classification and Identification of Ancient Indian Temple Pillars Era Using Deep Learning." International Journal of Intelligent SystemsApplications in Engineering, Vol. 12, No. 4, pp. 5436-5446, 2024.

[2] Fu Q., Hou M., Hua W. "Based on Improved PointNet Automatic Classification Method of Grotto Temple Statues." ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, Volume X-M-1-2023, 29th CIPA Symposium Paper, pp. 87-2023 (the numbering denotes article 87 in that issue), 2023.

[3] Rao A., Sindhu C., Suhail A., Mehta A., Dube Saksham. "Panchadeva: Sculpture Image Classification using CNN-SVM." Journal of Population Therapeutics and Clinical Pharmacology, Vol. 30, No. 9, pp. 332-344, 2023.

[4] Sridhar M., et al. 3DITA – A 3D benchmark dataset for Nagara-style Indian temple architecture, ISPRS Archives of the Photogrammetry, Remote Sensingand Spatial Information Sciences, XLVIII-M-9-2025, —, 1435–1441. 2025.

[5] Sasithradevi A., Sabarinathan S., Chanthini B., Subbulakshmi T., Prakash P. MonuNet: a high performance deep learning network for Kolkata heritage image classification, npj Heritage Science, 12, —, Article 242. 2024.

[6] Bahrami M., Albadvi A. "Deep Learning for Identifying Iran's Cultural Heritage Buildings in Need of Conservation Using Image Classification and Grad-CAM."Journal on Computing and Cultural Heritage,Vol.16,Issue 1,pp.—.2023

[7] Tan G., Zhu J., Chen Z. "Deep Learning Based Identification and Interpretability Research of Traditional Village Heritage Value Elements:A Case Study inHubei Province." npj Heritage Science, Vol. 12, Article 200. 2024

[8] Shih N.J. "How Can AI Help Us Connect to the Past in Terms of Conservation? AI - Assisted 3DRe construction of Temple Art: a case of Gong fan Palace, Yunlin,Taiwan." Technologies, Vol. 13,Issue 6, Article 229. 2025

[9] KhanS.,NaseerM.,HayatM.,ZamirS.W.,ShahbazK."Transformers in Vision:A Survey." IEEE Transactions on Pattern Analysis and Machine Intelligence, Vol. 45, No. 1, pp. 17-38, 2023.

[10] Huynh T., Nguyen H., Pham T., "Deep Learning for Hyperspectral Image Classification:A Critical Evaluation via Mutation Testing." IEEE Transactions on Image Processing, Vol. 33, No. 12, pp. 7654-7668, 2024.

[11] Lin K. X., Cho I., Walimbe A., Hollerer T., "Benefits of Synthetically Pretrained Depth-Prediction Networks for Indoor/Outdoor Image Classification." IEEE Access, Vol. 11, No. 4, pp. xxx-xxx, 2023.

[12] Dosovitskiy A., Beyer L., Kolesnikov A., Weissenborn D., Zhai X., Unterthiner T., Dehghani M., Minderer M., Heigold G., Gelly S., Uszkoreit J., Houlsby N.An Image is Worth 16×16 Words: Transformers for Image Recognition at Scale, 2020.

[13] Papa L., Russo P., Amerini I., Zhou L. A survey on efficient vision transformers:algorithms, techniques, and performance benchmarking, 2023

[14] Huo Y., Jin K., Cai J., Xiong H., Pang J. Vision Transformer (ViT)-based Applications in Image Classification, 2023.

[15] (Variant) Liu Z., Lin Y., Cao Y., Hu H., Wei Y., Zhang Z., Lin S., Guo B., "Swin Transformer: Hierarchical Vision Transformer using Shifted Windows", 2021.

[16] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, StephenLin, Baining Guo. "Swin Transformer:Hierarchical Vision Transformer using Shifted Windows." Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), pp. 10012-10022. 2021.

[17] Yingzi Huo, Kai Jin, Jiahong Cai, Huixuan Xiong, Jiacheng Pang. "Vision Transformer (ViT)-based Applications in Image Classification." 2023 IEEE 9th International Conference on Big Data Security on Cloud (BigDataSecurity),IEEE International Conference on High Performance and Smart Computing (HPSC),IEEE International Conference on Intelligent Data and Security(IDS), pp.33-38.2023.DOI:10.1109/BigDataSecurity-HPSC-IDS58521.2023.00033.

[18] Qiming Zhang, Yufei Xu, Jing Zhang, Dacheng Tao. "ViTAEv2:Vision Transformer Advanced by Exploring Inductive Bias for Image Recognition and Beyond." arXiv pre-print, 2022.